

Conservation technology

Biologia della Conservazione 25/26

Giuliano Colosimo

University of Rome Tor Vergata

- ▶ Conservation technology is defined as: “devices, software platforms, computing resources, algorithms, and biotechnology methods that can cater to the needs of the conservation community” (Speaker et al., 2022)

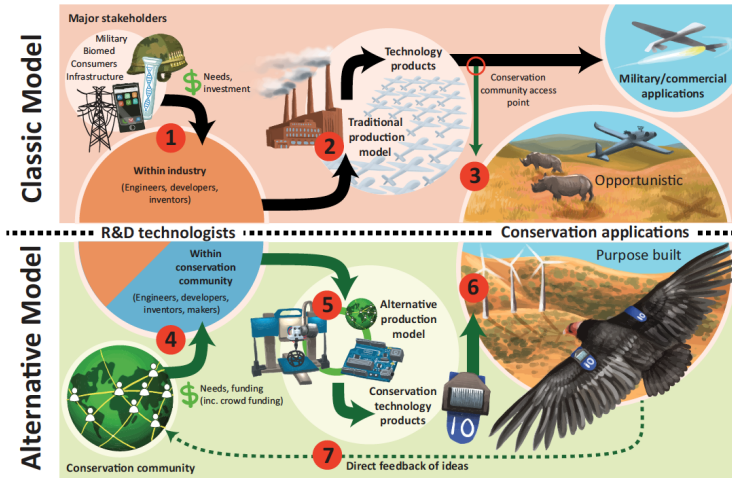


Figure 1: After Berger-Tal & Lahoz-Monfort (2018).

INTERFACE

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Perspective



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Conservation tools: the next generation of engineering–biology collaborations

Andrew K. Schulz^{1,2,†}, Cassie Shriver^{3,†}, Suzanne Stathatos^{5,†}, Benjamin Seleb^{3,†}, Emily G. Weigel³, Young-Hui Chang³, M. Saad Bhamla⁴, David L. Hu^{2,3} and Joseph R. Mendelson III^{3,6}

¹Haptic Intelligence Department, Max Planck Institute for Intelligent Systems, Stuttgart 70569, Germany

²Schools of Mechanical Engineering, ³Biological Sciences, and ⁴Chemical and Biomolecular Engineering, Georgia Institute of Technology, Atlanta, GA 30332, USA

⁵Division of Engineering and Applied Science, California Institute of Technology, Pasadena, CA 91125, USA

⁶Zoo Atlanta, Atlanta, GA 30315, USA

 AKS, 0000-0001-8007-5157; CS, 0000-0003-2970-1196; SS, 0000-0003-4351-4389; BS, 0000-0001-5023-1972; EGW, 0000-0001-5570-4142; Y-HC, 0000-0001-7987-6459; MSB, 0000-0002-9788-9920; DLH, 0000-0002-0017-7303; JRM, 0000-0001-5620-377X

The recent increase in public and academic interest in preserving biodiversity has led to the growth of the field of conservation technology. This field

Figure 2: After Schulz et al. (2023).

- ▶ Although no singular technology can solve the current global ecological crisis, devices such as camera traps, acoustic sensors, drones, biologgers, and satellites, as well as increasingly powerful genomic and artificial intelligence applications, hold the potential to empower conservationists to better understand and manage the socioecological systems in which they work (Speaker et al., 2022)

- ▶ Recent literature suggests that despite meaningful developments in this discipline, systemic issues such as unsustainable funding and development cycles, inadequate evaluation of solutions, and duplication of efforts are reducing its capacity to keep pace with escalating and emerging conservation challenges (Berger-Tal & Lahoz-Monfort, 2018; Speaker et al., 2022)

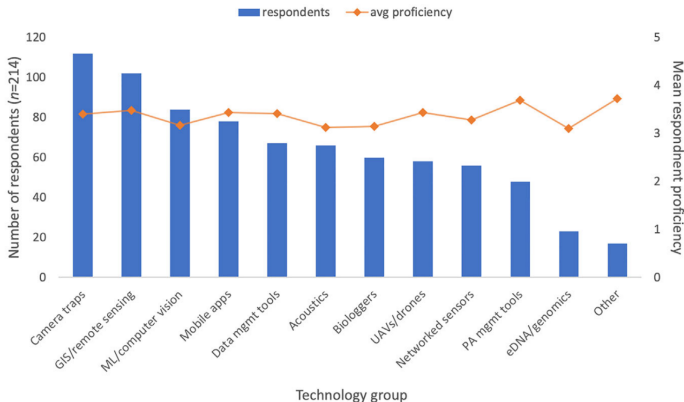


Figure 3: Conservation technologies frequently used by survey respondents and mean proficiency scores for each. After Speaker et al. (2022)

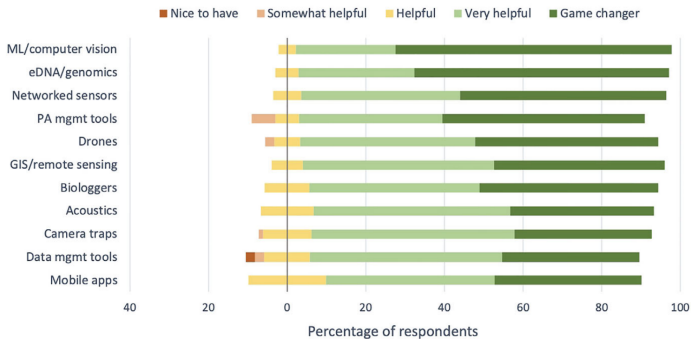


Figure 4: Capacity to advance conservation if current problems were addressed. After Speaker et al. (2022)

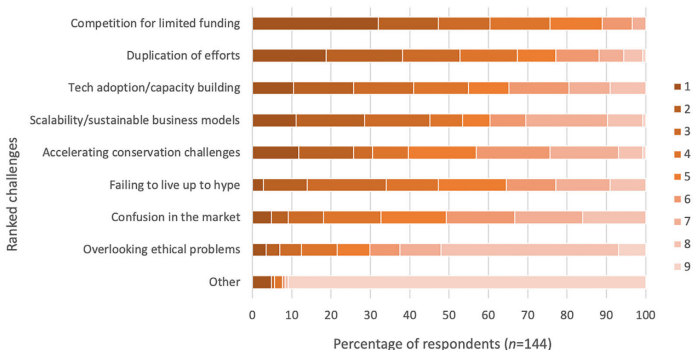


Figure 5: Survey respondent rankings of the greatest challenges facing the field of conservation technology (1, most important; 9, least important). After Speaker et al. (2022)

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